PERFORMANCE TESTING

Project Name: EduGenie

Project Type: AI-Powered Learning Assistant

Technology: Google Gemini Generative AI

1. INTRODUCTION

Performance testing is an important part of the EduGenie development process. It is used to evaluate how efficiently the application responds to user requests and communicates with the Google Gemini Generative AI service.

The main purpose of performance testing is to ensure that EduGenie provides a smooth and reliable experience when students use the application.

2. OBJECTIVES OF PERFORMANCE TESTING

The main objectives are:

- To check the response time of the application.
- To verify that user requests are processed correctly.
- To check the reliability of Gemini API communication.
- To identify delays or performance issues.
- To ensure that the application remains responsive during normal usage.
- To verify proper handling of errors and failed requests.

3. PERFORMANCE TESTING AREAS

The following areas are considered during testing:

3.1 Application Response Time

The time taken by EduGenie to receive user input, process the request, communicate with Gemini, and display the response is observed.

3.2 API Response

The communication between EduGenie and the Gemini API is checked to ensure that requests and responses are handled correctly.

3.3 User Interface Responsiveness

The application interface is checked to ensure that it remains usable while processing an AI request.

3.4 Error Handling

The system is tested under situations such as API failures, invalid input, or network interruptions.

3.5 Multiple Requests

The application can be tested with multiple requests to observe whether it continues to function correctly under repeated usage.

4. PERFORMANCE TEST CASES

Test Case 1: Normal User Request

Input:

A normal educational question or text.

Expected Result:

The application processes the request and displays an AI-generated response successfully.

Test Case 2: Large Educational Content

Input:

A relatively long educational passage.

Expected Result:

The application processes the content and returns an appropriate response without unexpected failure.

Test Case 3: Repeated Requests

Input:

Multiple learning requests submitted one after another.

Expected Result:

The application continues to process requests correctly.

Test Case 4: Invalid Input

Input:

Empty or invalid user input.

Expected Result:

The application provides an appropriate message instead of sending an invalid request.

Test Case 5: API Communication Failure

Input:

A request when the AI service cannot be reached.

Expected Result:

The application handles the failure gracefully and displays a suitable error message.

Test Case 6: Network Interruption

Input:

A request during an unstable or unavailable internet connection.

Expected Result:

The application handles the connection problem without crashing.

5. PERFORMANCE PARAMETERS

The following parameters are considered:

- Response Time
- API Request Success
- API Request Failure Handling
- Application Responsiveness
- Stability
- Resource Usage
- Error Recovery

6. PERFORMANCE TESTING PROCESS

The testing process follows these steps:

1. Start the EduGenie application.

2. Enter an educational question or content.

3. Submit the request.

4. Observe the time taken to receive the response.

5. Check whether the generated response is displayed correctly.

6. Repeat the process with different types of input.

7. Test error and network failure conditions.

8. Record and analyze the observed results.

7. EXPECTED PERFORMANCE

EduGenie is expected to:

- Respond to normal requests without unnecessary delays.
- Process user input correctly.
- Communicate reliably with the Gemini API.
- Display generated responses properly.
- Handle errors without application crashes.
- Remain usable during repeated requests.

8. PERFORMANCE IMPROVEMENT

Performance can be improved by:

- Optimizing application code.
- Avoiding unnecessary API requests.
- Keeping prompts clear and efficient.

• Handling errors and timeouts properly.

• Reducing unnecessary processing.

• Monitoring API response behavior.

• Improving the user interface during processing.

9. PERFORMANCE TESTING RESULT

Based on functional testing, the EduGenie workflow can be evaluated by checking whether user requests are successfully processed and AI-generated responses are displayed correctly.

Actual response time may vary depending on factors such as internet connectivity, server availability, API processing time, and request complexity.

10. CONCLUSION

Performance testing helps ensure that EduGenie provides a reliable and responsive learning experience.

By testing response time, API communication, repeated requests, error handling, and application stability, potential performance issues can be identified and improved.

The testing process provides a foundation for making EduGenie more efficient, reliable, and scalable for future development.